

SULUH: A Sovereign Educational AI System

A 35B-parameter educational AI trained, adapted, and served entirely on-premise — bound to four institutional harnesses under one PDPA-compliant governance layer. To our knowledge, the first university-built sovereign educational AI in Southeast Asia — named by the university's own motto, *Pengetahuan Suluh Budiman*.

VERSION	DATE	CLASSIFICATION	BUDGET	TIMELINE
3.4	21 July 2026	Internal R&D	RM 450K (SULUH AI)	12 months

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Demarcation

1 What SULUH is

SULUH is a sovereign educational AI initiative at Universiti Pendidikan Sultan Idris (UPSI). It comprises two separate but interdependent projects, developed and budgeted independently, but operationally inseparable.

PROJECT	WHAT IT IS	BUDGET	STATUS
SULUH AI	The sovereign model — a 35B-parameter educational AI trained, adapted, and served entirely on-premise	RM 450,000 RM 200K hardware + RM 250K manpower	R&D
SULUH Ecosystem	The institutional harness layer — four production applications consuming SULUH AI as their inference backend	Separate operational IT budget	Deployment-ready code exists

35B MoE teacher · 3B active per token	896 GB Aggregate cluster memory · 5 nodes	4 Institutional harnesses	10,000 Federated swarm devices	Month 14 Breakeven vs. cloud alternatives
RM 0 Software licensing cost				

Key value propositions

Sovereign by design No raw student PII ever leaves UPSI — the hardest PDPA questions never arise (§2)	Provable, not plausible IsRAG attaches provenance and a trust score to every answer (§7)	Owned economics RM 0 licensing · breakeven Month 14 · RM 30K/yr steady state (§9)	A research engine A 10,000-device evaluation testbed and five planned publications (§6, §11)
First-mover position The first university-owned educational sovereign AI in Southeast Asia (§2.4)			

Why this document exists. SULUH AI is a research novelty needing peer review and reproducibility. SULUH Ecosystem is institutional infrastructure needing operational budgets, not research grants.

This document — SULUH-SWARM — is the bridge that shows how the research connects to the operations in a single narrative for UPSI leadership and external reviewers.

2 The imperative: why now

Three forces converged within twenty-four months: students adopted AI faster than any technology in university history, universities became the world's most-attacked sector, and Malaysian data-protection law grew teeth. Together they make a foreign cloud the wrong foundation for institutional AI — and make 2026 the window to act.

2.1 Adoption has already happened — ungoverned

The HEPI/Kortext Student Generative AI Survey 2026 (n = 1,054) reports that **95% of students now use AI** in some form and **94% use generative AI on assessed work** — up from 66% any-AI use in 2024 — while only **38% of institutions provide AI tools**. The gap is shadow AI: coursework, drafts, and personal data flowing daily into foreign consumer chatbots under terms of service no one has reviewed. Doing nothing is not a neutral option; it is an unmanaged data-export programme. SULUH replaces shadow AI with a governed alternative that is better for students — bilingual, curriculum-aware, and free.

2.2 The threat landscape

Education is now the most-attacked sector globally — **4,388 cyberattacks per organisation per week** (Check Point, Q2 2025) — and 3.9 million education records were exposed in confirmed 2025 ransomware attacks, up 27% year-on-year (Comparitech). The largest breaches arrived through third-party vendors, not campus firewalls: University of Phoenix (3.5M people via one vendor zero-day, three months of dwell time), Columbia University (870K records, 460 GB, two months undetected after a single phished phone call), and PowerSchool (62M students via one compromised support credential without MFA). Microsoft observed that 39% of identity-based attacks targeted research and academic institutions. Every foreign AI vendor added to UPSI's stack multiplies this surface — while UPSI retains 100% of the liability.

2.3 The regulatory shift — PDPA (Amendment) Act 2024

OBLIGATION	IN FORCE	EXPOSURE
Breach of any data-protection principle	1 Apr 2025	Fine up to RM 1,000,000 and/or 3 years' imprisonment (previously RM 300K / 2 years)
Personal liability of directors & officers	2025	Named officers personally liable unless due diligence is proven
Mandatory breach notification	1 Jun 2025	72 hours to the Commissioner; failure alone up to RM 250,000 and/or 2 years
Mandatory Data Protection Officer	1 Jun 2025	Required at UPSI's data scale; registration within 21 days
Cross-border transfers	2025	Whitelist removed — lawful only to jurisdictions with substantially similar protection, evidenced per data flow

The sovereign inversion. Every cloud AI query carrying a student record is a cross-border transfer UPSI must justify, a vendor contract UPSI must audit, and a breach surface UPSI cannot see into. When data never leaves campus, the hardest compliance questions never arise. SULUH does not manage this risk — it removes it. (Average Malaysian breach cost, incl. fines, legal and churn: est. RM 3.2M.)

2.4 The regional landscape

PROGRAMME	BACKER	INVESTMENT	FOCUS
NMLP / SEA-LION · Singapore	IMDA · AISG · A*STAR	S\$70M ≈ RM 230M	National, general-purpose regional LLM
ILMU · Malaysia	YTL Corp × Universiti Malaya	Corporate · 500MW DC	National Bahasa Malaysia model, built from scratch
Sahabat-AI · Indonesia	GoTo Group	Corporate	Consumer & government services ecosystem
SULUH · Malaysia (UPSI)	A public university	RM 450K	Education-domain, university-owned, on-campus — first of its kind in SEA

Nations are spending tens of millions on sovereign AI; over 100 countries committed to AI sovereignty in the Bangkok Declaration (Feb 2026). Yet the higher-education seat remains unclaimed: to our knowledge, **SULUH is the first higher-education-focused sovereign AI built and owned by a university in Southeast Asia** — at roughly 0.2% of Singapore's national programme budget. National programmes supply the upside, not the competition: SULUH is designed to plug into NAI0/MyDIGITAL compute for its Phase-2 100B vision.

2.5 Why UPSI

UPSI has been first before: the first teachers' college (SITC, 1922), Malaysia's first education university (1997), and an early national adopter of open-source learning platforms (MyGuru, 2006). The torch is already in the university's motto — *Pengetahuan Suluh Budiman*. The first mover defines the national standard for educational AI, hosts the partnerships with ministries, MQA and 20 public universities, and attracts the talent that builds — and stays. Every year of delay costs RM 180–300K in cloud spend, twelve more months of shadow-AI exposure, and the first-mover title.

Sources: HEPI/Kortext Student Generative AI Survey 2026 · Comparitech Education Ransomware Roundup 2025 · Microsoft Digital Defense Report 2025 · Personal Data Protection (Amendment) Act 2024 & JPDP Guidelines (Feb 2025) · IMDA/AISG announcements · Fulcrum (ISEAS) 2026 · Digital in Asia 2026 · Simply Data Malaysia Threat Report 2025.

3 Verified model lineup

All models below were verified live on Ollama and HuggingFace on 20 July 2026. No assumptions; every tag resolves to a fetchable artifact.

TIER	MODEL	TAG	ARCH.	SIZE	MEM (Q4)	PLACEMENT
Cloud teacher	Kimi K3	API · Moonshot	Dense	100B+	—	API only
Cloud backup	Claude Opus 8	API · Anthropic	Dense	100B+	—	API only
Sovereign teacher	Qwen3.6-35B-A3B	qwen3.6:35b	MoE + Vision	35B / 3B active	~20 GB	Mac Studio
Production SLM	Gemma 4:31b	gemma4:31b	Dense	31B	~18 GB	DGX Spark
Agent SLM	Qwen3.5:9b	qwen3.5:9b	Dense · MM	9B	~6 GB (Q8)	Mac Studio / Spark
Embedding	bge-m3	bge-m3:latest	Dense	567M	~1 GB	All nodes
Federated student	Qwen3.5:0.8b / 2b	qwen3.5:0.8b	Dense	0.8–2B	0.5–1.2 GB	Campus laptops
Cluster peak (teacher + production + agent + embedding + KV cache + headroom)					~150 GB	of 896 GB — 6x headroom

Why 35B, not 70B or 100B

A 70B dense model at FP16 requires ~140 GB of weights plus 100+ GB of KV cache — around 250 GB total — which exceeds a single 256 GB Mac Studio when OS overhead and concurrent models are included. A **35B MoE achieves comparable quality with 3B active parameters per token**, fitting comfortably in 60–100 GB and leaving headroom for QLoRA jobs, evaluation, and burst load. Full-scale 100B pre-training is positioned as a **Phase-2 vision** contingent on national compute partnership (NAIO / MyDIGITAL), not a Year-1 deliverable.

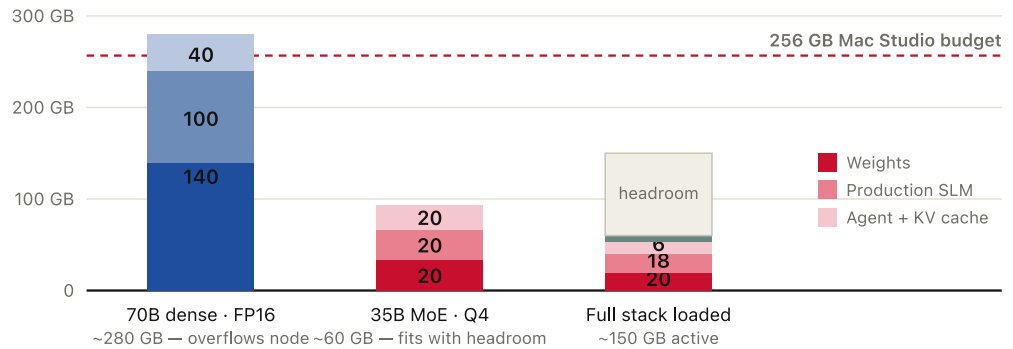


Figure 3.1: Memory footprint — 70B dense vs. 35B MoE serving. A 70B dense model overflows a 256 GB Mac Studio once weights, KV cache, and concurrent activations are included. The 35B MoE serves the full stack (teacher + production + agent + embedding) in ~150 GB with 6× headroom on the cluster.

4 Hardware & infrastructure

Cluster specification

COMPONENT	SPECIFICATION	QTY	UNIT (RM)	TOTAL (RM)
Teacher nodes	Mac Studio M3 Ultra · 256 GB unified · 2 TB SSD	2	68,000	136,000
Edge nodes	NVIDIA DGX Spark · GB10 Superchip · 128 GB coherent	3	14,000	42,000
Network	10GbE managed L3 switch, NICs, TB5 bridge cabling	1	8,000	8,000
Cabinet	12U wall-mount + cantilever shelves + UPS (30-min)	1	14,000	14,000
Hardware total				RM 200,000

Form factor. All five nodes are desktop/tower units (Mac Studio ≈ 20 cm cylinder; DGX Spark ≈ mini-tower). They mount on ventilated cantilever shelves in a 12U wall cabinet — 42U rails would be overkill. **Memory math:** 2 × 256 GB + 3 × 128 GB = **896 GB aggregate, not coherently pooled.** EXO disaggregates model layers across nodes; no single node holds the full 35B at FP16.

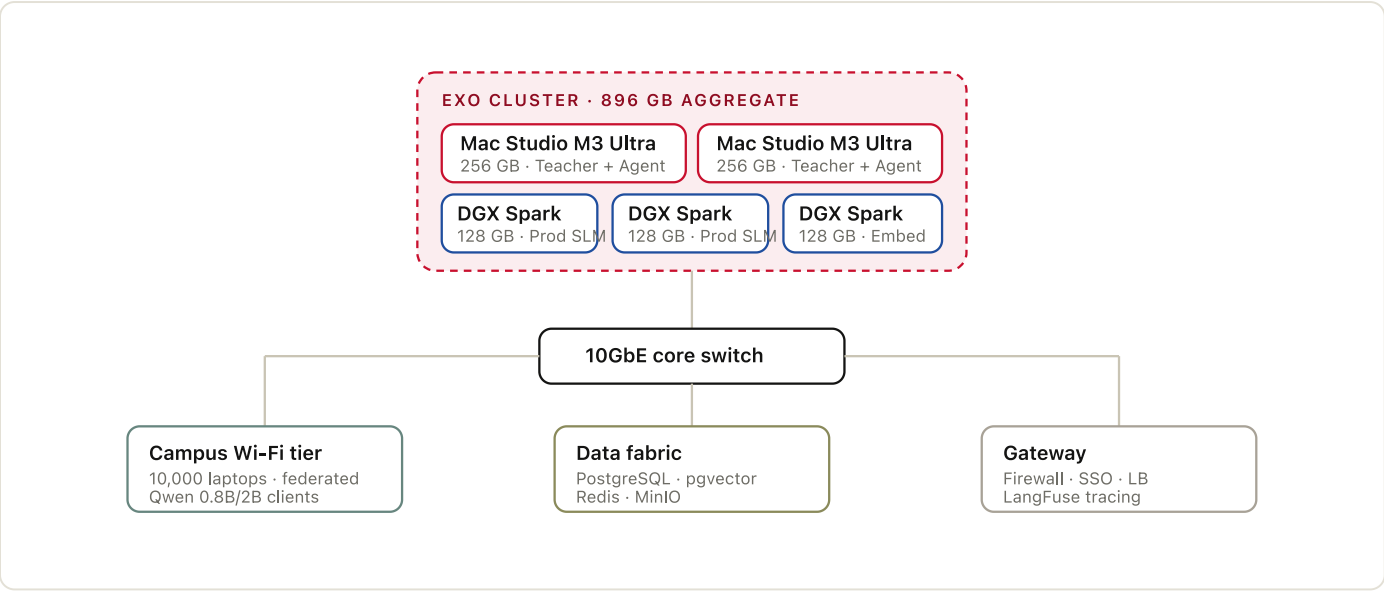


Figure 4.1: Compute topology. The EXO cluster aggregates ~896 GB across five nodes for teacher serving, production inference, and QLoRA jobs. The laptop swarm connects over campus Wi-Fi and exchanges only LoRA adapter deltas — never raw student data — with the aggregation service.

Software stack

LAYER	TECHNOLOGY	LICENSE
Orchestration	EXO (layer parallelism) + vLLM (inference)	Open source
Agent framework	LangGraph v1.0	Open source
RAG	pgvector + bge-m3	Open source
Document parsing	Docling	Open source
Observability	LangFuse	Open source
Frontend / API	Vue 3 + FastAPI	Open source
Databases	PostgreSQL 16, Redis 7, MinIO	Open source
Total licensing cost		RM 0

5 Three-track model strategy

SULUH runs three parallel tracks matched to what each tier of infrastructure can verifiably do — from anonymised cloud teachers down to campus laptops.

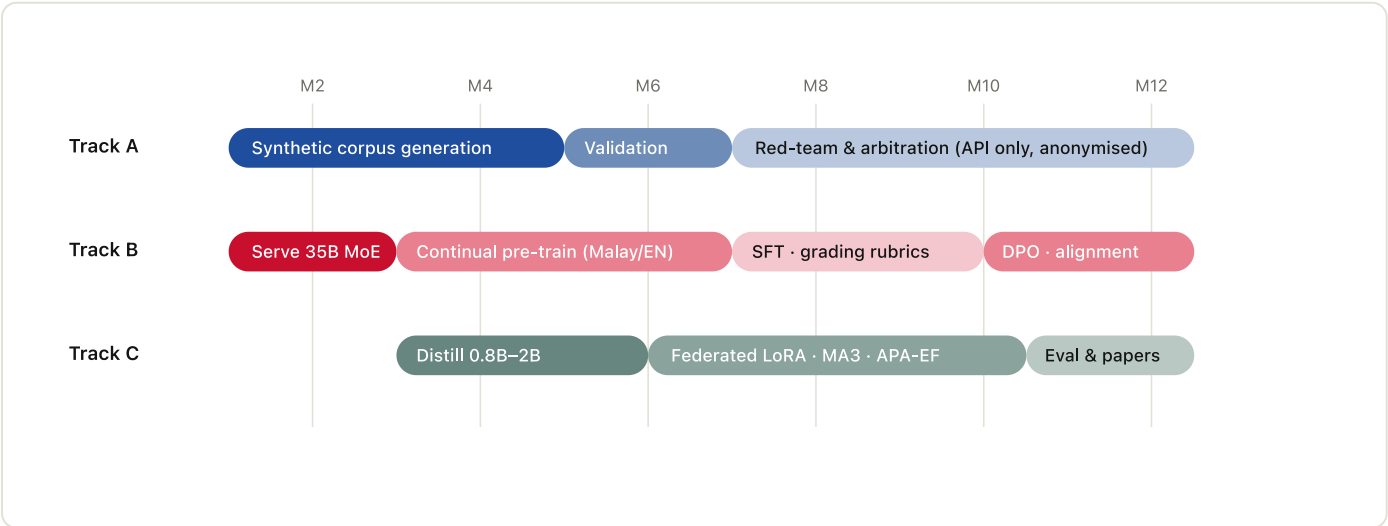


Figure 5.1: Three parallel tracks. Cloud teachers touch only anonymised curriculum data; the sovereign 35B is served on-prem from month 2 and continually adapted through DPO; the federated swarm trains what it can genuinely hold (0.8–2B distilled students) under differential privacy.

Track A — Cloud teacher (API, no PII)

Models: Kimi K3 (Moonshot), Claude Opus 8 (Anthropic). **Role:** generate curriculum-grounded synthetic training data from anonymised rubrics and model answers, and arbitrate quality on golden-set evaluation. **Data flow:** anonymised curriculum → cloud API → synthetic corpus → downloaded to cluster for local adaptation. **Cost:** ~RM 3,000–5,000 over 12 months. **Critical rule:** no raw student PII or assessment data ever leaves UPSI.

Track B — Sovereign teacher (on-premise, 35B MoE)

Model: Qwen3.6-35B-A3B (Apache 2.0, 262K–1M context). **Hardware:** Mac Studio 256 GB via EXO layer parallelism. **Pipeline:** Stage 1 synthetic corpus (M1–2) → Stage 2 continual pre-training on Malay/English (M2–4) → Stage 3 instruction tuning on grading & advising (M4–6) → Stage 4 federated adaptation on faculty-local documents (M6–9) → Stage 5 DPO alignment (M9–10) → Stage 6 accreditation-readiness evaluation (M10–12).

Track C — Federated student (campus laptops, 0.8–2B)

Models: Qwen3.5-0.8B and Qwen3.5-2B. **Role:** edge-deployed tutoring, local adaptation, federated research. **Training:** distilled from the 35B teacher via SFT; federated LoRA across 10,000 campus laptops under APA-EF differential privacy. Only adapter deltas are transmitted — never raw student data.

Value arrives early. The sovereign teacher serves real workloads from Month 2 — grading assistance and curriculum Q&A run on the base 35B while adaptation continues in the background. Funders do not wait twelve months for first value; the finish line is refinement, not activation.

6 Algorithmic contributions

The swarm is not just infrastructure — it is the project's primary research engine. Because laptops train models they can genuinely hold (0.8–2B), each contribution can be evaluated end-to-end on real heterogeneous hardware at a scale — 10,000 devices — few labs can replicate.

CONTRIBUTION	WHAT IT DOES	WHERE IT RUNS
MA3 Model-Agnostic Adaptive Aggregation	Weights adapter updates by device capability; handles the variance between Mac Studios and campus laptops in one aggregation round	Aggregation service on cluster
APA-EF Adaptive Privacy-Aware Elastic Federated	Differential privacy with elastic compression on LoRA deltas; top-k sparsification cuts communication by 60–80%; secure aggregation hides individual updates from the server	Laptop clients → cluster
LES-GA Localized Elastic Search, Gradient-Aligned	Keeps faculty-local RAG indexes consistent with the global curriculum knowledge base without full-model retraining	Data fabric + edge caches

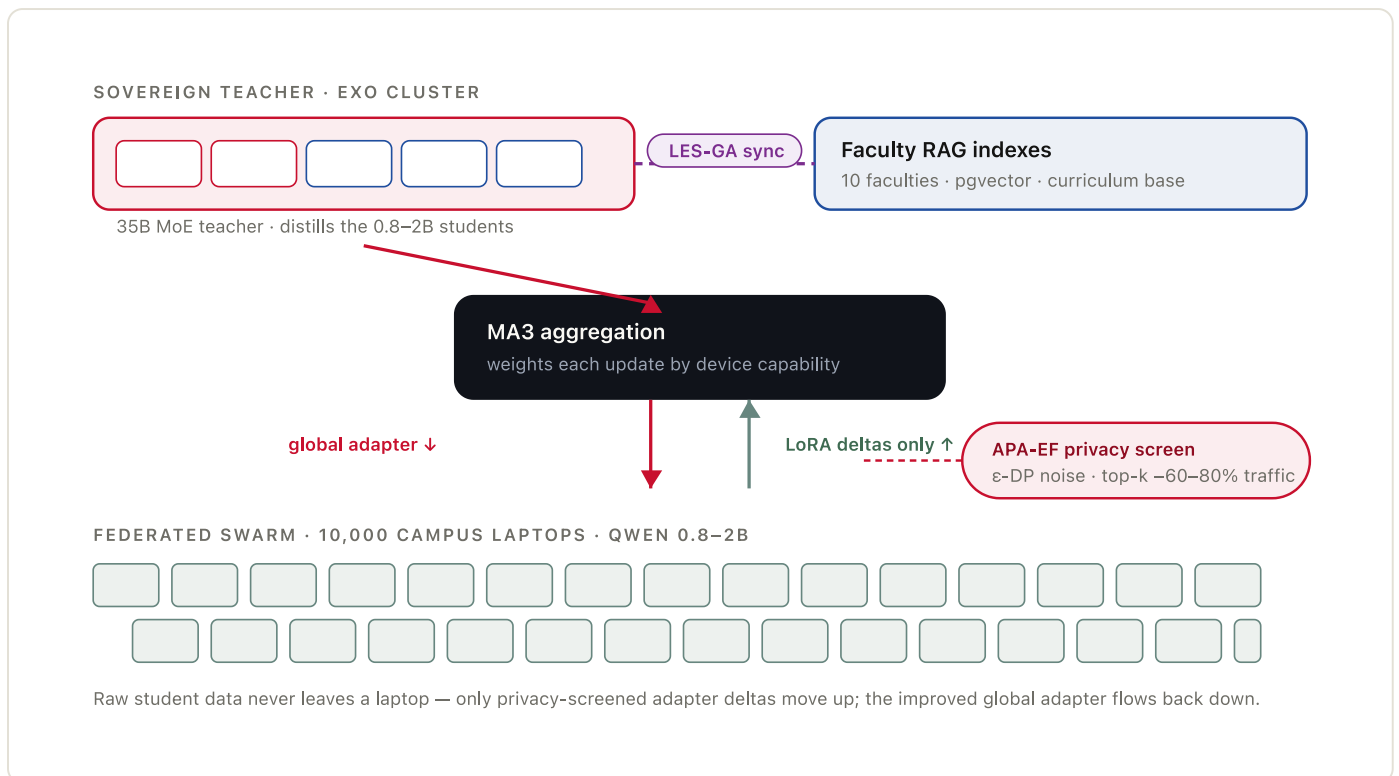


Figure 6.1: The swarm loop. Laptops train locally; APA-EF screens and compresses the updates; MA3 aggregates them into the sovereign teacher; LES-GA keeps faculty knowledge indexes aligned with the global curriculum base.

Foundational proof: Psyche

Psyche (Nous Research) has demonstrated that distributed training of transformer models is feasible across *untrusted internet participants*, using a coordinator/client consensus protocol with blockchain and TCP backend coordination. Psyche proves the core premise SULUH's federated track relies on: transformer models can be trained across many independent nodes without high-speed interconnects or a single coherent memory space.

SULUH operates under a **strictly easier trust assumption** — all 10,000 devices are institutionally owned — which enables APA-EF differential privacy (impossible in untrusted networks, where no node can be trusted with sparsification parameters) and LES-GA curriculum alignment (which requires a shared educational knowledge base only institutional ownership provides). SULUH's contribution is not proving distributed training works. Psyche proved that. SULUH's contribution is showing **how to deploy it safely, privately, and educationally within a sovereign institutional boundary**.

7 IsRAG: the trust engine

Trust — not fluency — is the binding constraint on educational AI. A model that writes beautifully but invents its sources cannot be allowed near a classroom. IsRAG is SULUH's answer: a retrieval architecture in which every claim carries its own chain of custody, and no answer reaches a student unverified.

7.1 The hallucination problem, measured

The failure mode is documented, large, and persistent. Walters & Wilder's 42-topic study found **55% of GPT-3.5 citations and 18% of GPT-4 citations entirely fabricated**, with substantive errors in 43% and 24% of the real ones. A 2025 Deakin University study (JMIR) found **56% of GPT-4o citations fake or erroneous** in academic literature reviews. Stanford's legal-hallucination studies measured 58–88% hallucination rates across major models. Crucially, fabrication is worst exactly where students most need help: **niche, low-resource topics** — which describes UPSI's Malay-language educational domain precisely. In education this is disqualifying: every false citation handed to a student is a defective teaching act with accreditation consequences.

7.2 Why retrieval alone is not enough

Standard RAG grounds answers in retrieved documents — but it treats a peer-reviewed journal article and an unverified blog post as retrieval equals, offers the user no signal about who created the knowledge or how it was verified, and still fabricates: recent measurements find even RAG-enabled systems hallucinate 3–13% of citation URLs at scale. Citations answer *where*; they never answer *whether to trust*. That missing layer — trust calibration — is the under-reported failure mode of production RAG.

7.3 The Isnad insight

Islamic scholarship solved this problem a millennium ago. The **Isnad (إسناد)** methodology validates a claim by its chain of transmission: every transmitter is named, rated, and contextualised, so a piece of knowledge is inseparable from the chain that produced it. IsRAG (*Isnad-chain Retrieval Augmented Generation*) ports this epistemology into the retrieval loop. Every knowledge unit

carries: **(1)** an immutable provenance chain recording every author, verifier, and adapter; **(2)** a trust score computed by a five-factor TrustEngine; and **(3)** a cultural-context tag drawn from a 13-framework registry that includes Malaysian and Islamic epistemologies — a first-class concern in a bilingual, multicultural deployment.

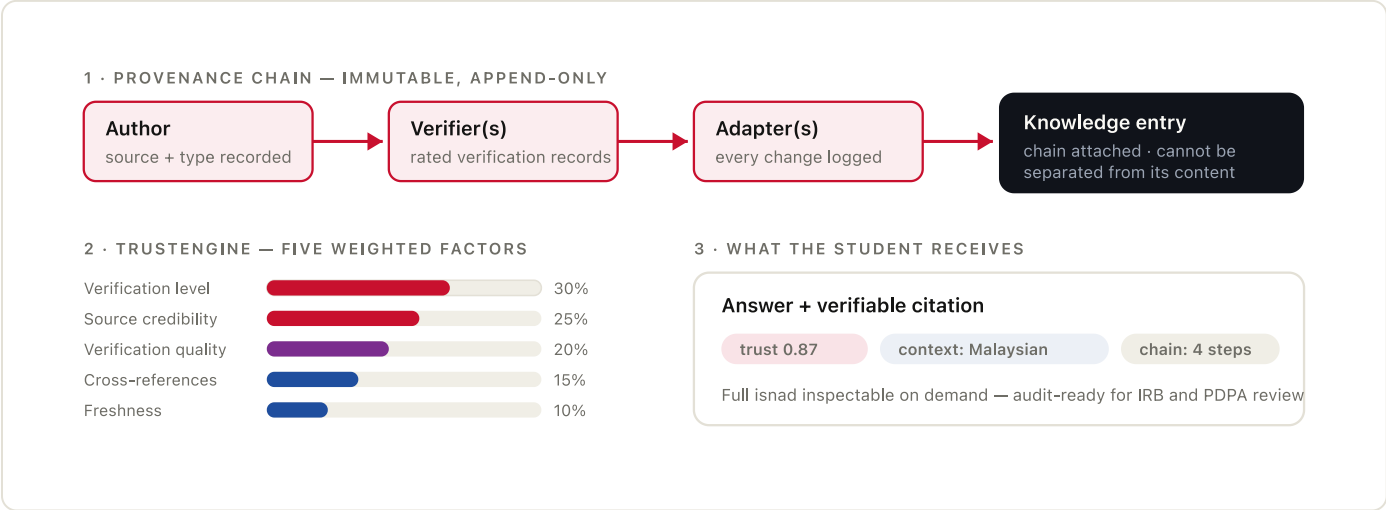


Figure 7.1: The IsRAG answer pipeline. Every retrieved knowledge unit carries an immutable provenance chain; the TrustEngine converts the chain into a calibrated 0–1 score; the student receives the answer with its citation, trust score, cultural-context tag, and inspectable chain.

7.4 What IsRAG buys SULUH

A citation tells you where; an isnad tells you whether to trust. The target is $\leq 5\%$ hallucination on the bilingual golden set — **up to 10× fewer false citations than the chatbots students use today** (18–56%, §7.1) — with a 100% auditable answer trail that is IRB- and PDPA-ready by design. No frontier model offers this: GPT, Claude, and Gemini answer; SULUH answers and proves. It is also the programme's clearest research contribution — Paper 1, targeting ACL/EMNLP/FAccT, evaluated against LangChain and LlamaIndex baselines on accuracy, trust calibration, cultural appropriateness, and hallucination rate — and a value proposition born naturally at a Malaysian university.

8 SULUH Ecosystem: the four harnesses

The Ecosystem is the "nervous system" that consumes SULUH AI as its brain. Seven original proposals merged into four harnesses along shared data dependencies, eliminating duplicate backends and letting all systems draw from one governed data fabric.

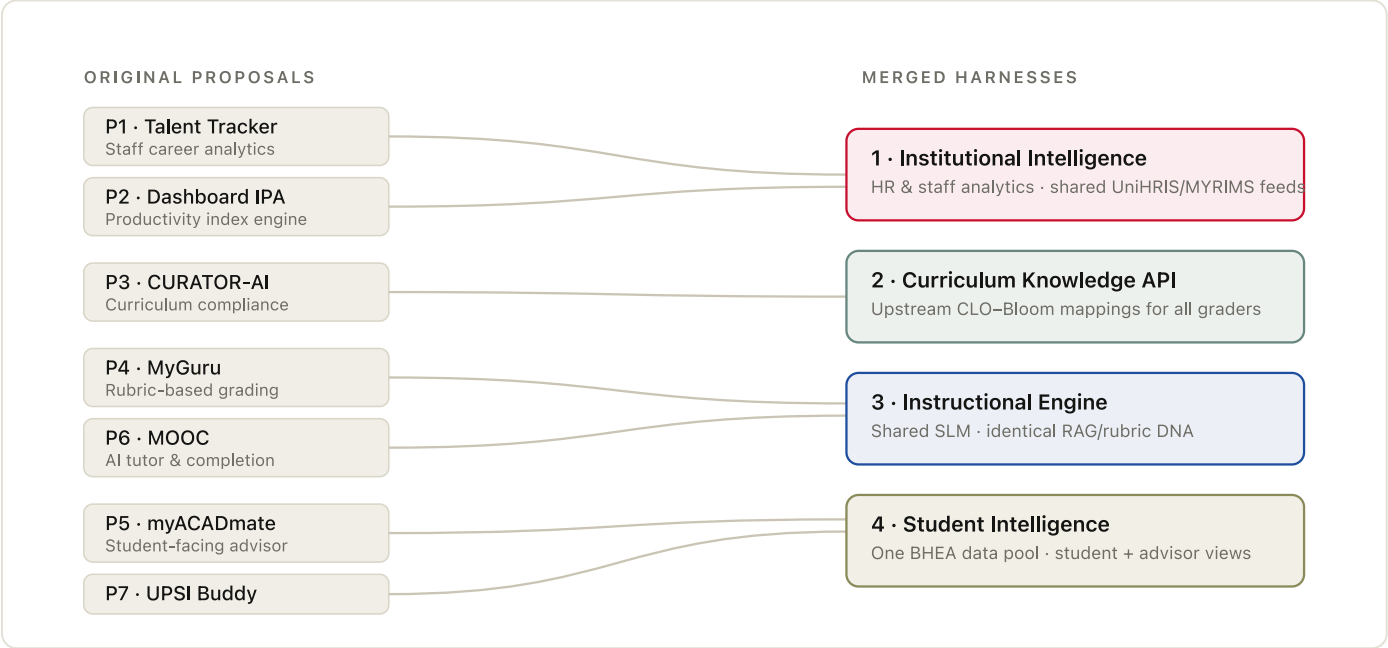


Figure 8.1: Seven proposals consolidate into four harnesses. *P1/P2 share HR databases; P4/P6 share grading and tutoring; P5/P7 share the student data pool; P3 becomes an upstream API consumed by Harnesses 3 and 4.*

#	HARNESS	OWNER	PRINCIPAL RESEARCHERS	DATA
1	Institutional Intelligence	BSM	Dr. Ahmad Amri, Dr. Nurul Hila	High
2	Curriculum Knowledge API	PPA	Dr. Amelia	Low
3	Instructional Engine	Pusat ICT / BHEA / BHEP	Dr. Mohd Muslim, Dr. Ahmad Wiraputra	Medium
4	Student Intelligence	BHEA / BHEP	Dr. Vasanthan, Pn. Nurul Ashikin, Jason Wong	High

Architecture layers

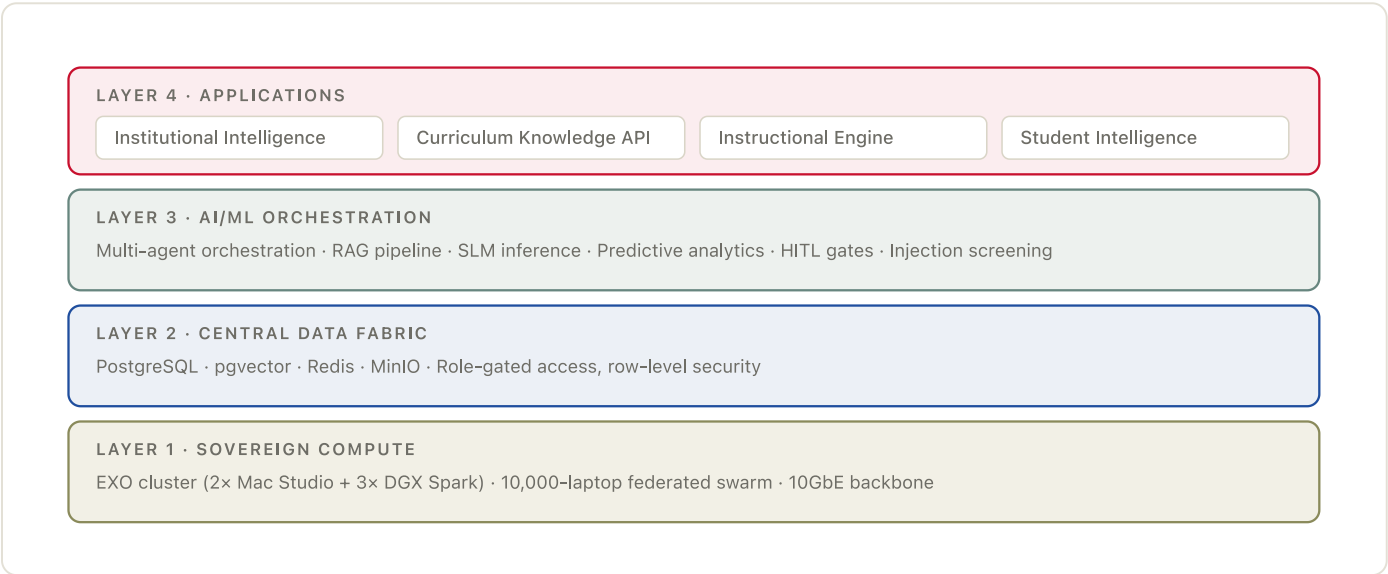


Figure 8.2: The four-layer stack. Governance is enforced at Layer 3 once for all applications; no harness bypasses the data fabric or the orchestration layer.

Governance & PDPA — enforced in code, not policy

GATE	RULE	ENFORCEMENT
Consent & purpose	Granular opt-in per harness; educational use only	<code>pdpa_gate.py</code> — rejects non-consented queries at API boundary
Data minimisation	7-year retention, 3-year auto-purge	PostgreSQL TTL + cron audit
Security	AES-256 at rest, TLS 1.3 in transit	Automated cert rotation
Federated privacy	ϵ -differential privacy on every swarm update	APA-EF runtime — no device identifier in gradient payload
Accountability	DPO Day 1, annual audit, 72-hour breach notification	IRB officer review + LangFuse trace logging

Mapped to the PDPA (Amendment) Act 2024. Gate 5 is the amendment's own checklist: a registered DPO (mandatory since June 2025), 72-hour breach notification, and audit-ready trace logging — the documented due diligence that shields UPSI's officers from the Act's personal-liability provisions (see §2.3).

9 Economic analysis

SULUH AI reaches breakeven against every cloud alternative within 14 months. After 36 months, sovereignty has saved between RM 240,000 and RM 600,000 — while every byte of student data stays on Malaysian infrastructure.

Budget (RM 450K total)

CATEGORY	ITEM	RM
Hardware	2× Mac Studio + 3× DGX Spark + network + 12U cabinet	200,000
Manpower	Research engineer (120K) + ML engineer (100K) + DevOps (80K), phased	250,000
Total (12 months)		RM 450,000

Annual OPEX (steady state)

Cloud API (synthetic data)	RM 5,000
Electricity (5 nodes, 24/7)	RM 8,000
Maintenance & warranty	RM 12,000
Network & backup	RM 5,000
Total OPEX	RM 30,000

Sovereign vs. cloud — cumulative 36-month cost

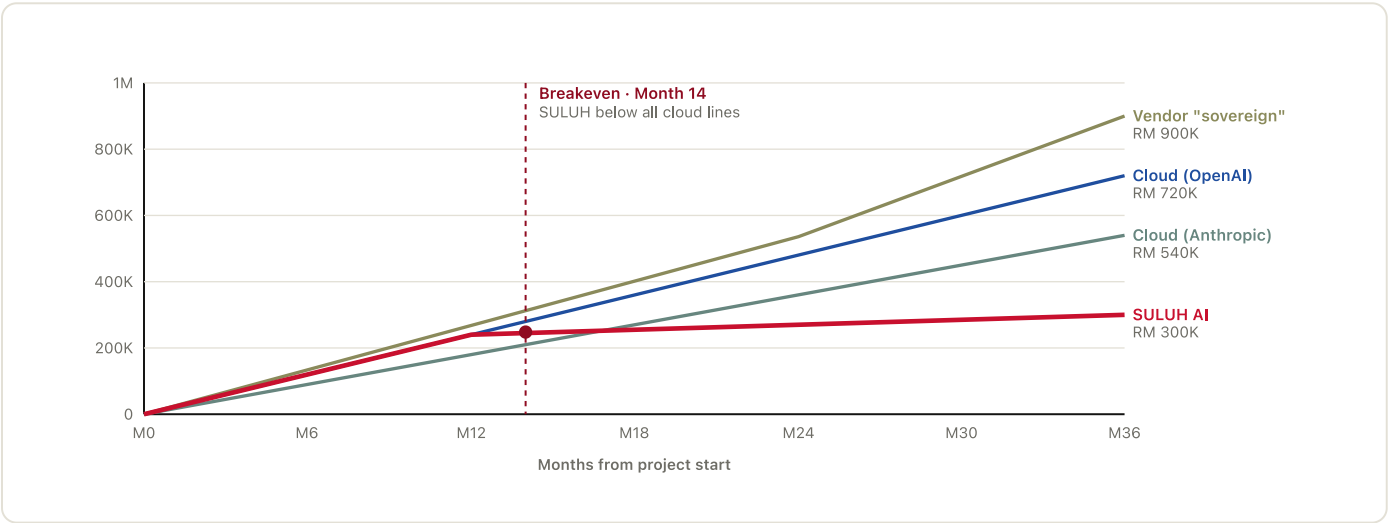


Figure 9.1: Cumulative cost over 36 months. *SULUH AI's cost curve flattens after Month 12 as the CAPEX-heavy build phase completes; cloud alternatives keep accumulating at RM 15K–25K per month. Breakeven against every cloud option is reached by Month 14, and by Month 36 sovereignty has saved between RM 240K and RM 600K.*

The unquantified line. These curves exclude the largest avoided cost: breach exposure. The average Malaysian data breach costs an estimated RM 3.2M in fines, legal fees and churn — and the 2025 amendments raised regulatory penalties to RM 1M per offence with personal liability for officers (see §2.3).

10 Evaluation plan

Golden set

200 query–answer pairs across five educational domains, bilingual Malay/English, authored by 30+ UPSI researchers and validated by subject-matter experts.

Baselines

MODEL	SIZE	DEPLOYMENT
SULUH AI (ours)	35B MoE	On-premise
Qwen3.6-35B-A3B (unadapted)	35B MoE	On-premise
Kimi K3	100B+	API
Claude 3.5 Sonnet	175B	API
Gemma 4:31b	31B dense	On-premise
GPT-4o (mini)	8B	API

Target metrics

METRIC	TARGET	MEASUREMENT
Grading accuracy (rubric adherence)	≥ 85% agreement	BERTScore + expert panel
Curriculum compliance (KPT/MQA)	≥ 90% correct CLO/PLO	Automated + auditor review
Student advising quality	≥ 80% satisfaction	Post-interaction survey
Malay language fluency	≥ 90% grammatical correctness	Native-speaker panel
Hallucination rate	≤ 5% on golden set	Citation verification + expert review
Latency	< 3 s grading · < 1 s chat	LangFuse tracing
PDPA compliance	100%	Audit log review
IRB gate accuracy	≥ 98%	IRB officer review

11 Publication plan

#	PAPER	VENUE	CONTENT
1	IsRAG	EMNLP / ACL	Isnad-chain provenance RAG with traceable citations
2	Dewan Council	EMNLP	Multi-agent deliberation with confidence-gated routing
3	SULUH AI	ACL / EMNLP / Nature MI	Sovereign 35B educational model, three-track architecture, economic analysis
4	Federated Swarm	NeurIPS / ICLR workshop	MA3, APA-EF, LES-GA evaluated at 10,000-device scale
5	Deployment Study	LAK / AIED	Full-platform evaluation across the four harnesses at UPSI

Paper 1 leads the programme. IsRAG is SULUH's flagship contribution and clearest differentiator against every frontier model. The full evidence base, architecture, and evaluation design are set out in §7.

12 Risk register

RISK	LIKELIHOOD	IMPACT	MITIGATION
SSO integration complexity	Medium	High	Phase 0 critical path, Week 1
Bilingual model underperformance	Medium	High	Phase 0 bench: Qwen3.6 vs Gemma4 vs Qwen3.5; swap paths preserved
Multi-agent consensus produces misleading outputs	Medium	High	Confidence-gated display, IRB gate, regression golden set
IRB gate adds latency	Medium	Medium	Async pre-screening, <500 ms target benchmarked in Phase 0
Hardware delivery delays	Medium	High	Mac Studio 7–10 business days; DGX Spark 2–4 weeks
PI bandwidth split across programmes	High	High	Engineering team absorbs implementation; PI focuses on architecture + papers
Federated swarm attrition	Medium	Medium	Incentive design, gamification, academic credit
Cloud API cost overrun	Low	Medium	RM 5K buffer, rate limiting, caching layer
PDPA non-compliance	Low	Critical	Privacy-by-design, DPO Day 1, annual audit
Phase-2 compute never materialises	Medium	Medium	35B is already production-viable; 100B is upside, not dependency

13 Demarcation: SULUH AI vs. Ecosystem vs. SWARM

TERM	DEFINITION	BUDGET	AUDIENCE
SULUH AI	The sovereign 35B model and its training pipeline	RM 450K (200K hw + 250K manpower)	Researchers, peer reviewers, grant bodies
SULUH Ecosystem	The four institutional harnesses + routing/orchestration layer	Separate operational IT budget	UPSI leadership: BSM, PPA, BHEA, Pusat ICT
SULUH-SWARM	This planning document — the bridge that hosts both narratives	N/A	Research Management, external auditors, partners

Why separate them. SULUH AI is a research novelty — a sovereign educational model needing peer review, reproducibility, and academic publication. SULUH Ecosystem is institutional infrastructure needing operational budgets, SSO integration, HR system connectors — not research grants. SULUH-SWARM is the bridge document that shows how the research (AI) connects to the operations (Ecosystem) in a single narrative for leadership.

The decision

Approve RM 450,000, and in twelve months UPSI owns Southeast Asia's first university-built sovereign educational AI — sovereign by design, provable by IsRAG, breakeven by Month 14, and shielded from RM 1 million liability by governance enforced in code. Every byte of student data stays home.

THE INVESTMENT

RM 450K · 12 mo

RM 200K hardware + RM 250K team, phased

THE RETURN

RM 240–600K saved

by Month 36 — plus breach exposure removed

THE UPSIDE

5 papers · 4 systems

live at UPSI · first-mover seat · Phase-2 to 100B

"SULUH: Menerangi masa depan pendidikan Malaysia dengan kecerdasan buatan berdaulat."

Document owner: SULUH Architecture Team · Distribution: UPSI Research Management, META, ICT, BSM, BHEA, BHEP, PPA, Pusat ICT · Next review: 19 Aug 2026

v3.4 supersedes v3.3 (21 Jul 2026): new §2 "The imperative: why now" (adoption data, threat landscape, PDPA (Amendment) Act 2024, regional positioning), new §7 "IsRAG: the trust engine", key value propositions, swarm-loop and IsRAG figures, and a closing decision block added, governance mapped to the 2024 amendment, economics annotated with breach exposure, palette aligned to UPSI corporate red.

v3.3 supersedes v3.2 (20 Jul 2026): Psyche foundational proof integrated, three-track model strategy re-anchored (Cloud/Sovereign/Federated), cost analysis reframed as sovereign-vs-cloud breakeven, budget aligned to RM 450K, all model tags verified live 20 Jul 2026.